

Research articles from 28 journals and 18 conferences were included in this review paper. Out of 28 journals referred 8 journals are published by Springer link, 16 journals are published by Science Direct (Elsevier), 2 journals are published by IEEE Access and 2 journals published by PubMed. Out of 137 articles selected 13 articles are from Springer Link, 37 are from Science Direct, 5 are from IEEE Transactions, 2 from PubMed and 80 papers are from different conference proceedings. Year wise number of publications of articles regarding diagnosis and classification of cancers that are included in this review paper are given in figure 4. It can be seen that from years 2008 to 2014 there is no significant change in number of publications but from 2014 onward there is

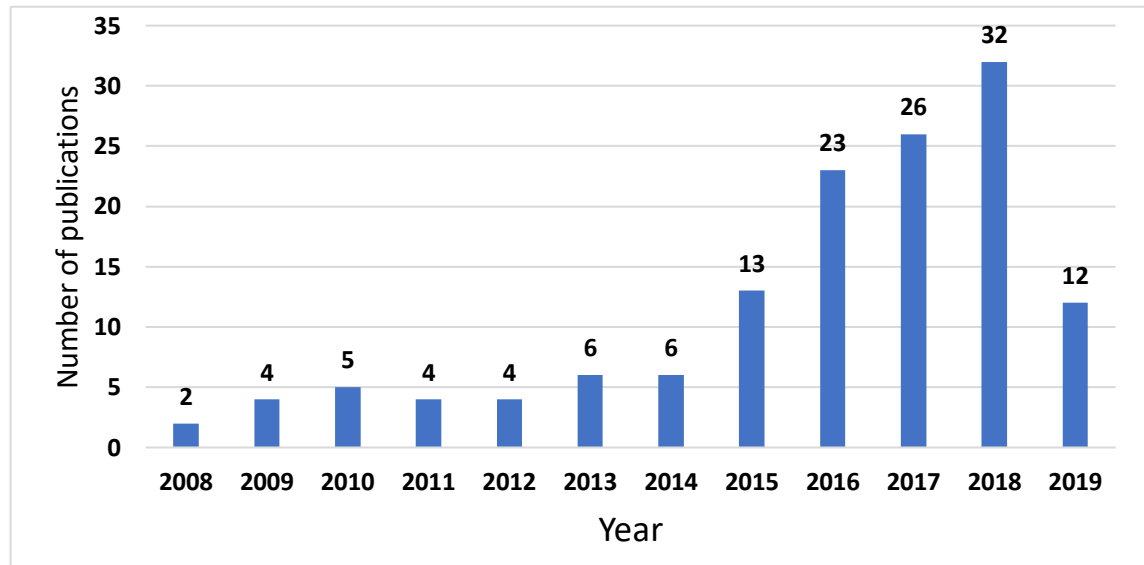


Fig 4: Showing year wise number of articles selected on cancer diagnosis and classification.

significant rise in the number of publications on the topic of cancer diagnosis.

2.3. Quality Evaluation Check (QEC):

Papers shortlisted from various electronic databases were screened against our QEC formulated by 2 independent reviewers, as shown below:

Q1: Is paper focused on classification of considered cancers using machine learning techniques?

Q2: Have author used standard imaging modalities for classification purpose?

Q3: Is research methodology explained?

Q4: Experimentation and results are properly explained?

Q5: Is result and conclusion consistent with data used?

Inter rated scale (Kappa =4) was used for selection of a paper by 2 reviewers. If the Kappa value of a research article is 4 or greater than 4 according to the QEC formulated above, then it is included in the data collection otherwise it is rejected by the reviewers. A consensus was reached by discussion method among author and reviewers for final selection and rejection of papers considered in this review paper. Initially a database of 3058 research articles was created based on the keywords and search strings shown in table 1. After going through screening process 1376 articles were selected for further consideration and 1682 articles were excluded because either they were not from artificial intelligence domain or they covered other cancer types. After a careful investigation of selected 1376 research articles, we found that many of them either did not use machine learning techniques or they did not used medical imaging data in their research. A fair share of articles was also found to be redundant, author published it in different journals. The number of such articles was 57. We also found some articles were published in other languages, so we excluded those articles also. Hence a further filtering of articles was done from which we selected 382 articles and rejected 994 articles. Out of these 382 selected articles 115 were covered breast cancer, 76 were related to brain cancer, 58 were on cervical cancer, 36 on oral cancer and 97 on skin cancer. These articles were further subjected to QEC with the help of our two reviewers. In our QEC we